

Status	Finished
Started	Wednesday, 6 May 2026, 2:22 PM
Completed	Wednesday, 6 May 2026, 3:10 PM
Duration	47 mins 45 secs
Marks	8.00/9.00
Grade	8.89 out of 10.00 (88.89%)

Question 1

Correct

Mark 1.00 out of 1.00

Select all the statements that are TRUE.

Select one or more:

- | | |
|---|--|
| ☒ a. Probabilistic Roadmaps (PRMs) are typically divided into a learning phase and a query phase. | ☒ Correct. PRMs build a roadmap first, then query it for paths. |
| ☒ b. Potential field methods can suffer from getting stuck even when a valid path to the goal exists. | ☒ Correct. Local minima are a well-known issue with potential field methods. |
| ☒ c. In configuration space, obstacles are transformed into regions that the robot must avoid. | ☒ Correct. Obstacles in workspace map to forbidden regions in configuration space. |
| ☐ d. Visibility graphs tend to produce paths that maximise clearance from obstacles. | |
| ☐ e. Exact cell decomposition methods are always more computationally efficient than approximate methods in high-dimensional spaces. | |
| ☒ f. Rapidly-exploring Random Trees (RRTs) are generally better suited to single-query problems than Probabilistic Roadmaps (PRMs). | ☒ Correct. RRTs are typically used for single-query planning, whereas PRMs are multi-query planners. |

This question tests understanding of configuration space, roadmap vs tree-based planners, and decomposition methods.

The correct answers are: In configuration space, obstacles are transformed into regions that the robot must avoid., Rapidly-exploring Random Trees (RRTs) are generally better suited to single-query problems than Probabilistic Roadmaps (PRMs)., Potential field methods can suffer from getting stuck even when a valid path to the goal exists., Probabilistic Roadmaps (PRMs) are typically divided into a learning phase and a query phase.

Question 2

Correct

Mark 1.00 out of 1.00

A mobile robot is navigating through a room with a point obstacle and a point goal, using a potential function. The positions are defined as follows:

- Robot start: (2, 1)
- Goal q_{goal} : (6, 6)
- Obstacle q_{obs} : (5, 4)

The attractive potential is $U_g(q) = (q - q_{goal})^2$, and the repulsive potential is $U_{obs}(q) = 1/(q - q_{obs})^2$. The total potential is given by

$$U(q) = U_g(q) + U_{obs}(q).$$

If the robot can move a distance of 1 in any direction, where would it be after taking one step using the potential function (i.e. taking one step of unit length in the direction of steepest descent of U)? Give your answer to at least 3 decimal places.

New x -coordinate:

Your last answer was interpreted as follows:

2.625

New y -coordinate:

Your last answer was interpreted as follows:

1.781

Correct answer, well done.

Correct answer, well done.

Worked solution

Treating $(q - q_{goal})^2$ as $|q - q_{goal}|^2$ and similarly for the obstacle term, the total potential at $q = (x, y)$ is

$$U(q) = (x - x_g)^2 + (y - y_g)^2 + \frac{1}{(x - x_o)^2 + (y - y_o)^2}.$$

The gradient is

$$\nabla U = 2(q - q_{goal}) - \frac{2(q - q_{obs})}{|q - q_{obs}|^4}.$$

Substituting the start position $q_s = (2, 1)$:

- $|q_s - q_{obs}|^2 = 18$
- $\nabla U_g = 2(q_s - q_{goal}) = (-8, -10)$
- $\nabla U_{obs} = -\frac{2(q_s - q_{obs})}{|q_s - q_{obs}|^4} = (\frac{1}{54}, \frac{1}{54})$
- $\nabla U = (-7.9815, -9.9815)$
- $|\nabla U| = 12.7802$

The robot steps a distance of 1 in the direction of steepest descent, $-\nabla U/|\nabla U|$:

$$q_{new} = q_s - \frac{\nabla U}{|\nabla U|} \approx (2.6245, 1.781).$$

The answer 2.62451832443, which can be typed as **2.62451832443**, would be correct.

The answer 1.78101015515, which can be typed as **1.78101015515**, would be correct.

Question **3**

Incorrect

Mark 0.00 out of 1.00

A mobile robot has paused at a waypoint and is using a noisy ultrasonic range sensor to estimate its distance to a fixed landmark, which it will use to refine its localization estimate. Because the robot is stationary while taking the readings, the true range is constant, so

$$x_{t+1} = x_t.$$

The noisy sensor readings are modelled as

$$z_t = x_t + \delta_t, \quad \delta_t \sim \mathcal{N}(0; 6),$$

i.e. zero-mean noise with variance 6 m^2 . The first three readings are

$$z_0 = 19 \text{ m}, \quad z_1 = 12 \text{ m}, \quad z_2 = 15 \text{ m}.$$

Using a Kalman filter with initial estimate $x_0 = 15 \text{ m}$ and $P_0 = 3 \text{ m}^2$, obtain the mean and variance of the robot's estimate of the range to the landmark after incorporating the last reading. Give your answers to at least 3 decimal places.

Posterior mean (m):

Your last answer was interpreted as follows:

14.25

Posterior variance (m^2):

Your last answer was interpreted as follows:

1.5

Incorrect.

The posterior *mean* is not correct (within tolerance 0.01 m).

The posterior *variance* is not correct (within tolerance 0.005 m^2).

Incorrect.

The posterior *mean* is not correct (within tolerance 0.01 m).

The posterior *variance* is not correct (within tolerance 0.005 m^2).

Worked solution

Because the state is constant ($x_{t+1} = x_t$) with zero process noise, the prediction step is the identity: $\hat{x}_t^- = \hat{x}_{t-1}^+$ and $P_t^- = P_{t-1}^+$. The filter therefore reduces to three measurement updates with

$$K_t = \frac{P_t^-}{P_t^- + R}, \quad \hat{x}_t^+ = \hat{x}_t^- + K_t(z_t - \hat{x}_t^-), \quad P_t^+ = (1 - K_t) P_t^-,$$

using $R = 6 \text{ m}^2$, $\hat{x}_0^- = 15 \text{ m}$, $P_0^- = 3 \text{ m}^2$.

Step 0 (incorporate $z_0 = 19 \text{ m}$):

- $K_0 = \frac{3}{3 + 6} = 0.3333$
- $\hat{x}_0^+ = 15 + K_0(19 - 15) = 16.3333 \text{ m}$
- $P_0^+ = (1 - K_0) \cdot 3 = 2.0000 \text{ m}^2$

Step 1 (incorporate $z_1 = 12 \text{ m}$):

- $K_1 = \frac{P_0^+}{P_0^+ + 6} = 0.2500$
- $\hat{x}_1^+ = \hat{x}_0^+ + K_1(12 - \hat{x}_0^+) = 15.2500 \text{ m}$
- $P_1^+ = (1 - K_1) \cdot P_0^+ = 1.5000 \text{ m}^2$

Step 2 (incorporate $z_2 = 15 \text{ m}$):

- $K_2 = \frac{P_1^+}{P_1^+ + 6} = 0.2000$
- $\hat{x}_2^+ = \hat{x}_1^+ + K_2(15 - \hat{x}_1^+) = 15.2000 \text{ m}$
- $P_2^+ = (1 - K_2) \cdot P_1^+ = 1.2000 \text{ m}^2$

So after the third reading the robot's estimate of the landmark range has mean 15.2000 m and variance 1.2000 m².

Sanity check: for a stationary scalar state the closed form gives $P_n^+ = R / (R/P_0 + n + 1)$ after $n + 1$ measurements, so here $P_2^+ = 6 / (6/3 + 3) = 1.2000 \text{ m}^2$, which matches.

The answer 15.2, which can be typed as **15.2**, would be correct.

The answer 1.2, which can be typed as **1.2**, would be correct.

Question 4

Correct

Mark 6.00 out of 6.00

For each scenario, select the most appropriate filtering method.

1. A robot moves along a track and maintains a single, continuously updated estimate of its position. Sensor readings are noisy but always cluster tightly around the true value, and the robot's motion is predictable.

Kalman Filter



2. A robot is placed in a building where many corridors look identical. After a few sensor readings, there are still several distinct, equally plausible locations where the robot might be.

Particle Filter



3. A drone estimates its height using smooth, high-frequency sensor readings. Its estimate evolves gradually over time, and uncertainty remains centred around a single best guess.

Kalman Filter



4. A robot uses a camera in an environment with repeated visual patterns (e.g., identical shelves). Observations can match multiple places, and the robot must keep track of several competing location hypotheses.

Particle Filter



5. A robotic system has a well-understood motion model and receives frequent, small corrections from sensors. Its belief about the state never splits into separate possibilities.

Kalman Filter



6. A robot is suddenly relocated without its knowledge and must re-estimate its position from scratch using sensor data, initially considering many widely separated possibilities.

Particle Filter



Kalman filters are appropriate when the system maintains a single, well-localised estimate that evolves smoothly over time. Particle filters are more suitable when multiple competing hypotheses

must be maintained, such as in ambiguous or global localisation scenarios.